

Title: Mastermind Rivals

Genre: Multiplayer Third-Person Shooter with RTS Elements

Mode: Online Battle Royale-Style Deathmatch (One-for-All)

1. Executive Summary

Overview

Each player controls a **Mastermind** and a **Shooter Player**. The **Mastermind** operates from a strategic RTS-like interface, managing the battlefield, deploying support, and defending their base. The **Shooter Player** engages in fast-paced third-person combat.

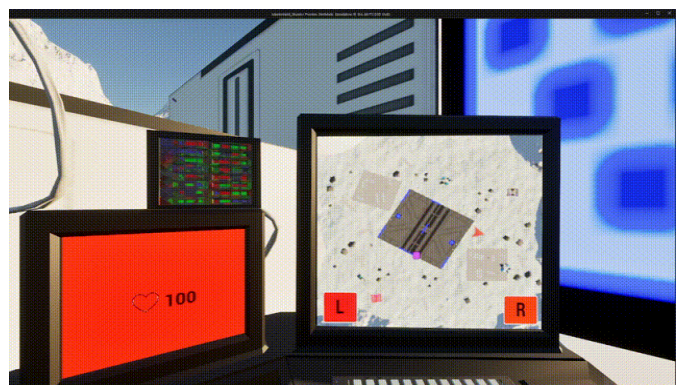
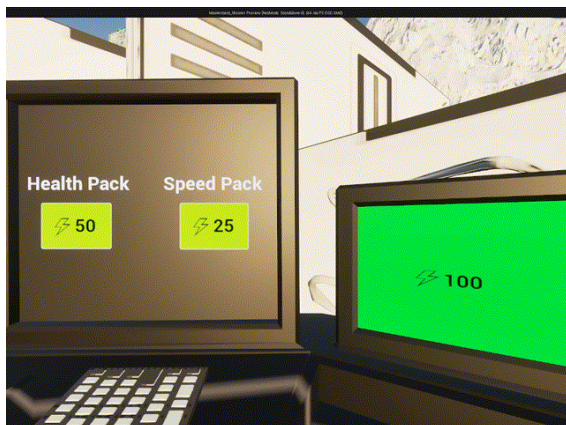
Objective

Players must find and eliminate rival **Masterminds** while defending their own base. The last surviving **Mastermind** wins the match.

Game World

- A futuristic battlefield filled with **strategic interactables**.
- Players start in **individual bases** with key resources.
- The world is populated with **hostile NPCs** (exploding enemies & turrets).

2. User Interface Mock-ups



3. Controls

Shooter Player Controls

- **WASD** - Movement
- **Mouse** - Aim
- **Left Click** - Shoot
- **Space** - Jump
- **Cntrl** - Crouch

Mastermind Controls

- **1, 2, 3** - Switch between cameras
- **Click** - Select/Interact with map objects

4. Design Patterns

Patterns Used & Implementation

State Pattern – Manages different movement states (Running, Walking, Shooting).

Observer Pattern – Multiplayer logic triggers different responses based on events.

Singleton Pattern – Ensures key managers (Game Session, UI Controller) are globally accessible.

Template Pattern – Objects like doors, interactables, and pickups derive from parent classes for easier customization.

Object Pooling – Optimizes turret bullet spawning and NPC explosion effects.

5. Tasks & Schedule Breakdown

Project Management & Tracking

Project progress is managed using Jira, and the development roadmap can be viewed here:

 [Mastermind Project - Jira Board](#)

Team Roles & Contributions

- **Felix**: Worked on the Shooter Player mechanics, animations, multiplayer environment, and menus.
- **Saul**: Developed Enemy AI, including turrets and exploding NPCs, and contributed to the music, FX, and map design.
- **Max**: Built the Mastermind mechanics, including pickup spawns, seamless switching between Mastermind and Shooter Player, and the interactive minimap system. Also assisted with map design.

6. UML Diagrams

